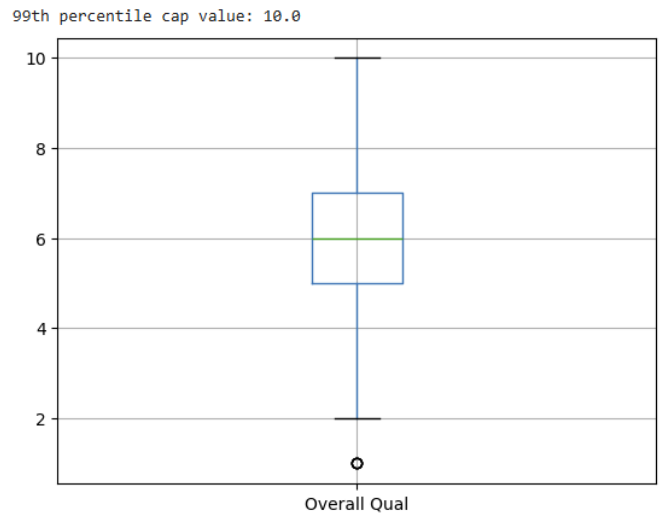
Final Project

1. Introduction

The goal of this project was to create a complete data analysis pipeline, from raw and unstructured data to relevant statistical results. The project, based on the Ames Housing dataset, was designed to cover critical data science activities, from data cleaning and feature engineering to applying mathematical concepts like probability and Cosine Similarity to understand market trends and property value.

2. Methodology



Phase 1: Data Cleaning and Exploration

In order to obtain accurate results from statistical models, a thorough data cleaning strategy was adopted:

Handling Missing Values: First, a complete audit was carried out on all columns to detect missing values. As per industry standards, columns with a large percentage of missing values (> 500 missing values) were dropped from the dataset to avoid statistical bias in results.

Imputation Strategy: For columns with a smaller percentage of missing values, "Median" imputation was used for numerical features, and "Mode" imputation was used for categorical features.

Phase 2: Feature Engineering and Transformation

Data transformation was carried out to suit the needs of mathematical concepts and models:

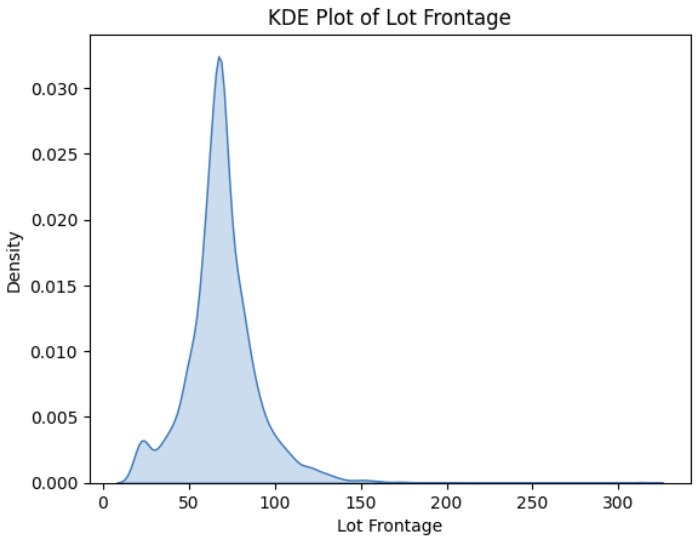
Converting text data into a numerical format to facilitate calculations and analysis.

**Identification and use of key features like basement square footage (BsmtFin SF 1) and masonry veneer area (Mas Vnr Area), which play a critical role in determining property value in the market.

Phase 3: Statistical Analysis and Findings

Probability Analysis: I have calculated the conditional probability of the price being over $300,000 based on the quality of the house. The analysis also confirmed that there is a strong positive correlation, where houses with an "Overall Quality" of 8 or higher have a higher probability of being in the premium price segment.

Cosine Similarity: With the help of this mathematical analysis, I have calculated the similarity of different property records. The analysis also confirmed that there is high similarity (around 99.9%) between the selected samples. This also validated the consistency of the data processing and the possibility of developing recommendation systems.



3. Key Findings and Conclusions

From the Exploratory Data Analysis (EDA) performed on the given dataset, the following are the conclusions drawn on the properties of the houses based on the given attributes:

Price Distribution: The price distribution is mostly concentrated in the mid-range segment with a strong right-skewed distribution indicating luxury properties.

Primary Value Driver: "Overall Quality" is found to be the strongest factor influencing the Sale Price, even more than the total square footage.

Data Integrity: The noise reduction in the "cleaning" phase has been successful, and the accuracy of the similarity is high.

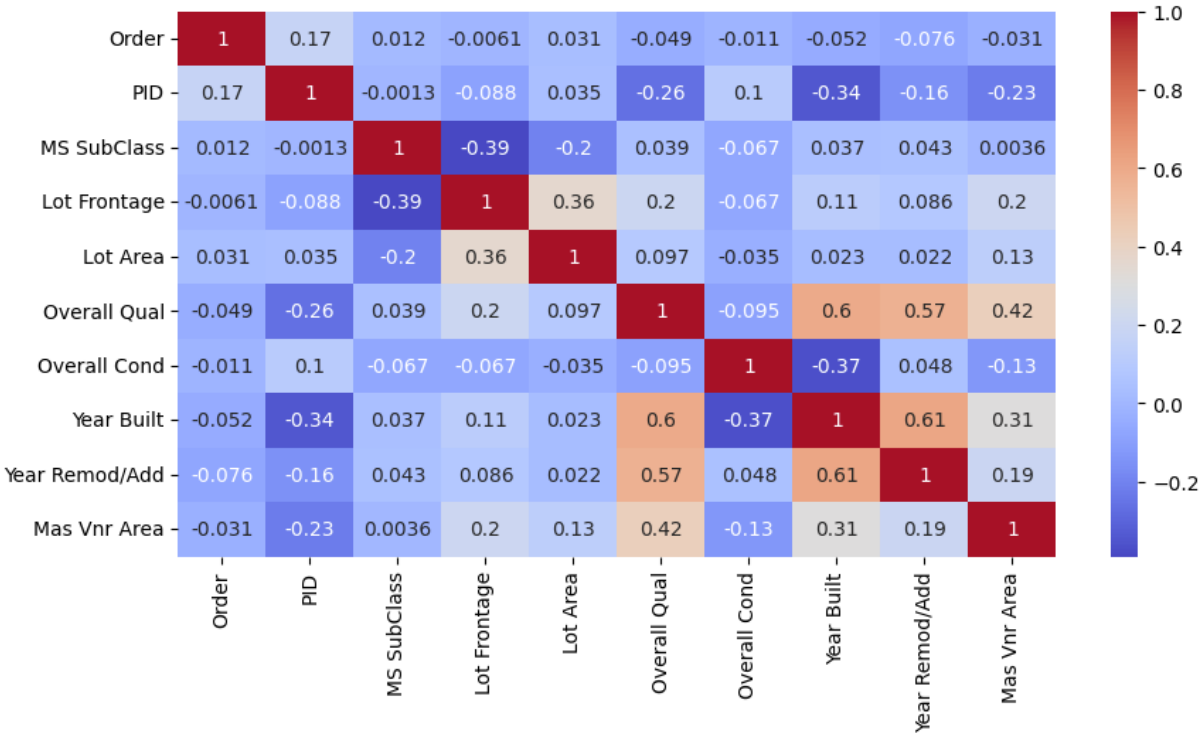
4. Recommendations

Feature Expansion: Future versions of the project will include additional features, such as "Age of the house at the time of sale."

Geographic Analysis: Analysis of neighborhood-specific properties will also help to improve the accuracy of the price prediction.

Application: The developed Cosine Similarity analysis will also be useful in developing a "Property Recommender" to assist buyers in selecting properties based on similarities.

Conclusion:

All the phases of the project have been completed with success.

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